

PKI & HTTPS Web Server Configuration (SEED Lab)

Asma Razick

Task 1: Becoming a Certificate Authority (CA)

The goal of the first task was to become a root certificate authority. In order to accomplish this, there were several steps I needed to take. The first step was to create and start the container Docker. This created the network net-10.9.0.0 so that the containers could communicate. This also built the web server and set up my environment. Next, I needed to generate an RSA private key to be used for generating my self-signed certificate as the CA. The private key is important because it is how the CA creates digital signatures for all the certificates it issues, including its own. After printing the contents of the certificate, I was able to obtain several key pieces of information. The validity period of my certificate is from November 8th, 2025, at 18:24:39 GMT until November 6th, 2035, at 18:24:39 GMT. The certificate is part of the certificate chain, because it is at the top, otherwise known as the root. I came to this conclusion because I noticed that the issuer is the same as the subject, meaning it was self-signed. Below is an image of the lines in the output I am referring to.

```
Issuer: CN = www.modelCA.com, O = Model CA LTD., C = US
Validity
  Not Before: Nov  8 18:24:39 2025 GMT
  Not After : Nov  6 18:24:39 2035 GMT
Subject: CN = www.modelCA.com, O = Model CA LTD., C = US
```

The key length of the private key used in generating the certificate is 4096 bits. Below is an image of the line that revealed this information.

RSA Private-Key: (4096 bit, 2 primes)

There are three X509v3 extensions present in the certificate. These extensions included the subject key identifier, the authority key identifier, and the basic constraints. Below is an image of the three extensions present in my certificate.

```
X509v3 extensions:
  X509v3 Subject Key Identifier:
    1D:16:0D:EF:49:EC:69:ED:BA:7B:CB:1F:AC:03:45:9E:85:ED:81:DA
  X509v3 Authority Key Identifier:
    keyid:1D:16:0D:EF:49:EC:69:ED:BA:7B:CB:1F:AC:03:45:9E:85:ED:81:DA

  X509v3 Basic Constraints: critical
```

The RSA algorithm used for generating the certificate and the key had a public exponent e , a private exponent d , a modulus n , and two secret numbers p and q , such that $n = pq$. The value of e was 65537 (0x10001). Below is an image of the private exponent, d , the modulus, n , the secret number p (prime 1), and the secret number q (prime 2).

privateExponent:

6c:98:b8:73:0a:98:a3:d7:eb:08:f5:71:96:5e:65:
c1:b2:eb:03:29:8c:a2:08:5e:27:d2:28:ef:d7:af:
2c:f0:cd:a8:45:54:72:4a:fe:02:ab:31:fe:6b:0b:
7e:25:61:6a:a8:a5:a8:42:d7:e5:e8:a3:31:55:3c:
0b:49:31:10:2f:0d:fa:f0:28:53:bf:1a:c5:65:69:
3d:c0:ec:f5:f8:5e:2a:54:0e:bf:91:22:3d:07:c2:
06:63:fb:b2:8d:8c:b9:9f:8f:ed:21:4e:8e:d1:ad:
96:8c:54:03:aa:ad:94:32:30:9d:ba:3c:42:a4:89:
83:f7:2b:6a:46:b3:ef:48:59:a3:24:d0:30:20:85:
c1:0c:6f:ad:39:f0:43:d8:08:3e:6b:4a:bc:aa:f9:
e1:c2:7d:79:87:7d:55:ba:b0:17:4b:f7:8e:ad:d5:
85:b5:bb:d0:28:54:d3:ae:9e:08:6b:8e:60:51:31:
0d:e4:84:7c:59:70:8e:37:bb:cb:21:3c:5b:76:34:
05:b9:ca:48:94:4d:9f:bd:d5:82:41:6f:1f:f7:f3:
c7:94:95:bf:84:65:f6:93:bd:f4:43:78:76:e8:21:
76:1a:60:1c:4b:63:df:91:f4:97:aa:6e:67:f4:c7:
28:77:c5:8b:fb:11:c5:9f:e9:bd:90:56:45:2e:d8:
96:ee:cc:f0:4a:19:ef:03:b8:5d:68:5d:77:9d:77:
7c:be:4b:2a:4a:8f:99:a7:f5:14:21:ae:02:db:fa:
38:a8:35:d5:02:81:3a:07:22:cc:26:c0:2f:de:d2:
50:d9:cc:13:3c:b1:1e:98:88:96:c8:9e:36:83:85:
c7:45:8b:dd:4b:85:25:08:f2:23:b3:43:12:50:a0:
73:dd:f5:96:de:5b:ff:8e:e3:a6:39:14:a3:04:e7:
44:3a:62:cf:2b:42:77:b9:fb:1f:bb:80:14:4e:72:
7d:79:c2:8a:a7:4a:ae:1c:ef:a5:bc:fe:de:24:f3:
58:50:d7:68:66:cf:7f:cd:b6:f0:47:99:f5:ae:dc:
da:d2:70:bc:34:5c:04:c4:5d:7c:3a:36:43:fd:0f:
24:85:c7:06:b0:e2:e8:80:7f:21:e4:03:d0:f5:59:
4c:c9:c3:46:d2:f5:a9:4c:8c:f5:32:85:c8:f3:de:
da:0e:fd:aa:bc:b4:91:04:46:af:b6:71:37:3d:dd:
3d:fe:8d:64:d8:4b:a6:80:94:7c:60:a2:3a:c5:ef:
bf:b6:16:45:8c:8d:ad:d6:74:b0:23:31:58:5f:6b:
5d:e0:30:e3:4c:f5:f4:39:cb:eb:8c:26:73:14:d6:
d0:54:45:62:d3:f2:e9:37:5b:a3:e0:ae:10:43:b2:
43:31

modulus:

00:a8:6e:95:d2:fa:da:bc:b3:33:c9:73:db:f0:5e:
7d:08:f3:d2:1d:1e:ce:56:18:c1:88:4c:88:f5:1a:
c7:4a:64:42:49:37:62:7a:57:4d:de:04:f5:b7:69:
f4:ad:8b:2d:a0:5f:ad:51:d8:c1:19:80:d6:f7:45:
b9:15:37:d7:c7:ba:d3:57:d4:8b:58:08:2e:9a:cb:
6c:2b:2c:6e:54:73:66:fa:d2:45:70:92:49:48:b8:
38:42:7c:4b:b8:c8:cc:0b:32:a0:b5:f4:79:94:f2:
d3:48:ae:ca:31:e0:9c:d5:9e:a5:43:14:0c:1f:4a:
6a:bd:ab:8d:b2:a5:78:44:2b:e4:9c:db:2a:a6:b2:
56:00:d3:39:f2:73:72:63:6b:c9:57:e6:38:e3:7e:
44:80:58:34:da:a4:b2:c1:b2:24:b8:5f:09:c0:80:
f3:09:c7:a7:4a:9c:59:b6:b3:92:24:6b:c2:b0:0b:
40:b7:88:40:39:91:25:db:71:ed:e7:0e:b9:99:53:
1f:fb:6e:48:38:11:88:ce:65:69:fb:28:b0:dd:ed:
7e:f6:d5:f4:d1:14:02:07:88:f1:45:9d:3d:4e:cb:
37:04:ab:8a:56:35:72:de:39:1b:a4:76:84:13:9f:
db:ec:63:6b:59:27:65:89:b0:47:6d:0f:07:af:6d:
e7:b2:cf:69:f5:f2:b5:81:0b:10:23:37:3b:6f:98:
b4:b5:5c:86:5c:10:ec:56:88:0e:40:7d:eb:60:be:
b3:d7:03:9f:dd:2e:e9:4f:46:0d:40:c0:4f:16:4c:
7e:70:47:9b:0c:da:48:39:8e:42:f9:96:2b:1d:5e:
33:44:80:86:4d:9b:b7:81:ef:42:c2:6e:e2:cd:57:
36:55:3a:08:8f:af:03:ba:90:93:d8:93:d1:ca:ec:
22:27:79:31:fd:3a:0a:07:eb:9a:74:4e:0e:1c:77:
3e:5b:34:a0:03:6a:e9:92:89:7c:88:9a:d9:dc:c8:
f7:94:c8:63:df:ce:72:b7:93:c3:93:22:af:fa:ed:
ac:53:86:8c:e3:1c:ce:c3:b3:f4:7c:55:d3:06:2d:
f7:01:e3:23:e3:73:1b:c0:24:d1:a0:b4:0d:e2:08:
ba:83:77:f7:ad:a9:c6:d3:c5:7e:72:0b:8c:3f:7b:
83:2a:b0:f2:d4:af:e8:1a:b7:f9:28:9c:11:64:42:
98:57:8b:29:07:a4:c4:b0:5a:05:2c:13:b8:e3:c6:
e6:eb:dd:6d:f3:a5:4e:6d:66:9d:99:ef:d7:08:58:
be:d3:37:fd:da:2d:01:af:b5:7e:83:a9:11:d0:23:
24:17:0e:ef:ef:31:21:57:37:09:42:8b:3a:54:c6:
9a:f5:6f

prime1:

00:d0:95:fb:6b:27:69:98:72:c8:fe:e6:9d:e5:aa:
b1:6d:1f:8b:f7:3e:eb:4f:30:1f:ad:83:70:eb:39:
19:ad:5e:09:47:76:07:05:bd:7d:a4:f9:f1:a9:66:
39:bf:dc:da:32:8f:72:1a:18:cc:c2:f8:7d:e0:f7:
99:8d:ca:07:46:15:3c:b6:55:54:d6:2a:7c:8c:b4:
36:0c:4c:c9:c0:5d:12:db:5b:0e:26:ec:77:37:21:
52:0d:b3:dc:e6:00:1b:c3:38:0b:04:1e:eb:54:c5:
b7:80:90:bd:78:81:38:12:40:66:6b:18:46:d5:57:
da:69:14:38:96:35:ce:d1:68:b6:79:18:7b:db:7a:
6e:0e:87:f5:31:97:d7:8b:4a:a5:48:2d:d0:d3:d9:
c1:5c:bb:87:53:12:23:b3:b2:4f:97:4c:75:b2:be:
4b:91:b6:74:c3:52:4c:55:3b:a3:7e:03:b4:f5:8e:
19:88:14:42:cb:3c:a3:46:c0:19:a8:0d:0b:93:b1:
b7:05:3d:c1:df:57:f5:ce:9b:27:33:63:ab:21:0c:
c4:31:ad:27:45:99:84:00:4c:54:63:d9:af:c4:6d:
cc:d0:71:ee:7f:1e:9d:cf:9f:5c:12:57:ba:01:11:
c3:72:37:15:79:38:a4:4f:df:7c:fa:d7:79:64:73:
28:79

prime2:

00:ce:b7:f8:48:15:d7:ba:1b:b1:d5:3e:fd:9a:fa:
fa:1d:a0:d0:20:4e:57:99:fa:f2:9d:20:5a:58:8d:
8c:18:e6:73:ea:ff:c7:d7:0f:20:f9:1f:df:98:4a:
8e:00:64:37:ad:d5:26:2c:7a:e0:ba:c3:96:c2:b4:
1f:8a:8a:94:05:0d:c3:1f:60:bd:a6:8a:6b:57:4d:
57:f6:48:a3:ac:06:e8:01:c8:01:7f:94:b7:f8:a0:
41:4a:f3:d6:b4:9c:db:b7:ec:17:66:21:ac:11:d2:
e7:a4:2f:b8:45:5d:30:f6:2c:4f:92:95:87:1f:25:
18:e3:9e:0a:c5:58:93:0f:a1:d0:03:e4:3f:cc:32:
c5:4b:f0:cc:9d:6a:a4:8b:ab:4f:04:c2:ba:e3:f7:
b1:31:d0:f1:b2:61:c7:06:39:1e:11:a6:f5:c3:ce:
26:27:61:fd:28:07:6a:2b:39:2b:5c:02:fc:53:ce:
7c:ba:7c:6c:b1:2b:cc:66:a3:fb:f0:e4:28:e6:52:
19:4c:b6:76:f3:15:94:3c:4f:18:74:dd:46:06:1f:
b8:63:e4:d6:13:12:69:43:8a:cb:08:92:a1:04:03:
02:2d:a8:d2:91:5f:d6:2b:4c:fa:f0:02:a7:39:ab:
7e:d5:2b:a0:95:cb:9f:91:9e:60:9c:2b:92:eb:71:
63:27

Task 2: Generating a Certificate Request for Your Web Server

The next task was to generate a certificate request for my web server. The first step was to generate a CSR. The name of my web server is www.razick.com. Below is a screenshot of the command for generating a CSR along with the output.

```
root@ac9a897b0cc0:/# openssl req -newkey rsa:2048 -sha256 \
> -keyout server.key -out server.csr \
> -subj "/CN=www.razick.com/O=Razick Inc./C=US" \
> -passout pass:dees
Generating a RSA private key
.....+++++
.....+++++
writing new private key to 'server.key'
-----
root@ac9a897b0cc0:/# openssl req -in server.csr -text -noout
Certificate Request:
  Data:
    Version: 1 (0x0)
    Subject: CN = www.razick.com, O = Razick Inc., C = US
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
      RSA Public-Key: (2048 bit)
      Modulus:
        00:dc:72:c7:b9:2c:a1:39:71:a8:b0:03:4a:ac:54:
        8e:3d:eb:4c:2f:67:cf:72:7f:9f:55:b5:ac:0d:88:
        32:56:2b:f4:81:77:de:1c:e6:0c:2a:c9:12:17:6d:
        dc:5a:b4:7a:71:48:2c:74:be:c5:8c:c5:5b:99:2f:
        f5:71:b7:15:34:23:6c:66:2e:cc:9e:c1:71:b8:e7:
        7d:7b:02:d4:bc:71:b5:e9:50:6d:53:2b:21:ba:c8:
        7e:c2:f5:81:d3:c2:19:f7:4b:ec:a4:be:c6:a0:91:
        90:73:22:28:3b:89:48:b2:7f:14:ed:83:ef:c5:86:
        54:c6:25:33:ac:15:e4:4c:55:8b:2b:2d:ae:d5:55:
        1f:4f:30:c5:7d:d9:a3:a3:21:e9:21:13:71:52:7d:
        cd:41:f5:9f:f1:5a:7f:5c:aa:68:7b:4f:6e:ae:fc:
        c6:5e:a5:68:09:0d:21:e4:e8:bd:2e:13:22:4b:8e:
        fb:88:99:6b:4d:aa:c9:e4:2d:2d:4e:3a:c4:c5:46:
        ca:4f:34:8f:d1:09:17:dd:ea:57:21:ba:3c:5c:8e:
        6d:a1:dc:05:15:8b:a0:ce:11:aa:c0:22:4b:18:a0:
        b8:63:8f:41:4b:4f:71:4e:58:ab:9e:82:05:de:d7:
        95:c3:36:61:02:94:1c:0e:40:42:ea:89:21:94:fa:
        a7:5d
      Exponent: 65537 (0x10001)
    Attributes:
      a0:00
  Signature Algorithm: sha256WithRSAEncryption
    c3:83:39:9a:b4:95:88:7e:79:6e:ff:ac:34:b0:7c:2f:3a:f4:
    98:35:44:a6:99:b1:8d:be:b3:9b:b1:4a:b0:d7:85:db:fa:ff:
    81:9d:22:c2:a9:5b:8a:54:74:df:51:54:f2:77:74:50:fe:fb:
    dc:e5:1b:5e:22:00:fb:f9:7b:1c:68:21:15:60:9d:ea:f6:d5:
    45:23:2a:7c:9f:a1:59:a1:9b:14:2d:08:b5:fa:de:3e:7c:2d:
    f7:60:82:f2:1d:9e:c7:8f:46:37:70:a1:71:06:7f:24:e0:82:
    b9:b6:33:ad:f5:03:31:97:30:98:4f:8f:c5:6e:64:66:0f:a7:
    5d:87:15:22:78:24:8e:ab:7b:f5:43:3a:33:1d:4b:8a:94:e6:
    b8:cc:ba:e3:3d:8e:a4:0a:b7:6c:e6:ba:70:16:a3:3a:fa:bc:
    39:5a:b2:50:61:02:14:5f:18:7c:33:05:07:e4:b6:6a:d4:0c:
    e2:e4:db:21:0f:ea:0f:7d:d5:5b:4d:1b:28:ad:2f:c0:b7:9b:
    8a:48:e9:6c:20:93:e2:a2:63:f7:c1:72:f3:95:5e:c1:8a:a4:
    66:a7:4e:0c:4d:24:a6:fb:7a:c8:ac:a0:05:36:6e:f1:72:c3:
    29:d6:ef:5c:60:84:8c:63:c6:57:fd:11:70:30:49:b0:44:7c:
    57:32:d8:70
```

Finally, I generated a CSR to allow my certificate to have multiple names, which are highlighted in the image below. Below is a screenshot of the command for generating the CSR along with the output.

```
root@ac9a897b0cc0:/# openssl req -newkey rsa:2048 -sha256 \
> -keyout server.key -out server.csr \
> -subj "/CN=www.razick.com/O=Razick INC./C=US" \
> -addext "subjectAltName = DNS:www.razick.com, \
> DNS:www.razickZ.com, \
> DNS:www.razickB.com, \
> DNS:www.razickH.com, \
> DNS:www.razickA.com" \
> -passout pass:dees
Generating a RSA private key
.....+++++
.....+++++
writing new private key to 'server.key'
-----
```

Task 3: Generating a Certificate for Your Server

The purpose of task 3 was to take the CSR from task 2 and generate a certificate for my server. Below is the screenshot of the command I used to generate and sign the certificate along with the output.

```
root@ac9a897b0cc0:/# openssl ca -config myCA_openssl.cnf -policy policy_anything \
> -md sha256 -days 3650 \
> -in server.csr -out server.crt -batch \
> -cert ca.crt -keyfile ca.key
Using configuration from myCA_openssl.cnf
Enter pass phrase for ca.key:
Check that the request matches the signature
Signature ok
Certificate Details:
    Serial Number: 4096 (0x1000)
    Validity
        Not Before: Nov  8 19:16:37 2025 GMT
        Not After : Nov  6 19:16:37 2035 GMT
    Subject:
        countryName           = US
        organizationName      = Razick INC.
        commonName            = www.razick.com
    X509v3 extensions:
        X509v3 Basic Constraints:
            CA:FALSE
        Netscape Comment:
            OpenSSL Generated Certificate
        X509v3 Subject Key Identifier:
            87:8C:DA:E7:A2:95:CA:92:75:27:FB:52:5A:EA:F6:B3:DF:D9:3B:BA
        X509v3 Authority Key Identifier:
            keyid:D3:59:53:11:94:40:F7:22:71:91:17:04:24:87:CE:CB:DF:18:17:E2

        X509v3 Subject Alternative Name:
            DNS:www.razick.com, DNS:www.razickZ.com, DNS:www.razickB.com, DNS:www.razickH.com, DNS:www.razickA.com
Certificate is to be certified until Nov  6 19:16:37 2035 GMT (3650 days)

Write out database with 1 new entries
Data Base Updated
```

Below is the output of the generated certificate.

```
root@ac9a897b0cc0:/# openssl x509 -in server.crt -text -noout
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number: 4096 (0x1000)
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: CN = www.razickCA.com, O = Razick CA LTD., C = US
    Validity
      Not Before: Nov  8 19:16:37 2025 GMT
      Not After : Nov  6 19:16:37 2035 GMT
    Subject: C = US, O = Razick Inc., CN = www.razick.com
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
      RSA Public-Key: (2048 bit)
      Modulus:
        00:97:72:e8:ab:62:3f:7e:b9:f0:77:0f:7d:5f:7e:
        be:d7:f3:02:bc:8b:1f:10:ec:0f:3e:c8:b8:3f:8a:
        1d:1b:5e:53:3e:c6:27:fc:7e:30:90:70:fc:43:58:
        c9:07:69:1e:7c:09:70:6c:36:71:89:37:84:0c:20:
        a3:a9:a3:f2:cd:d3:0d:4b:be:aa:3b:2a:7b:47:d9:
        15:b1:68:31:70:38:7e:69:e2:18:d1:18:20:ea:cf:
        75:a0:83:1b:5e:23:a6:2c:3c:c3:68:52:f6:5f:e2:
        7b:41:c1:fc:e0:a1:e0:aa:41:9e:f5:1c:03:b3:3e:
        82:76:8b:65:21:b4:c6:d0:6a:59:3c:dd:f6:48:4a:
        c3:38:86:59:2c:2d:71:76:39:39:80:f4:a3:99:8e:
        2d:c3:ae:f8:db:d1:46:c0:67:ce:c7:48:c9:e8:9d:
        a6:63:8e:43:70:f7:2b:c3:ea:dc:d8:5c:2e:81:88:
        4f:9d:46:06:21:73:9d:14:3d:d4:50:f4:4a:cc:1a:
        c7:e4:67:7e:99:af:85:06:06:5a:fa:86:bf:12:1f:
        c0:3c:dc:a1:8e:93:58:ab:6b:99:82:49:35:dc:89:
        90:87:cb:b9:1a:18:84:bd:28:2c:12:60:fe:5e:ce:
        78:dd:38:be:23:47:fa:58:55:97:12:c8:73:53:38:
        24:3f
      Exponent: 65537 (0x10001)
    X509v3 extensions:
      X509v3 Basic Constraints:
        CA:FALSE
      Netscape Comment:
        OpenSSL Generated Certificate
      X509v3 Subject Key Identifier:
        87:8C:DA:E7:A2:95:CA:92:75:27:FB:52:5A:EA:F6:B3:DF:D9:3B:BA
```

```
X509v3 Authority Key Identifier:
  keyid:D3:59:53:11:94:40:F7:22:71:91:17:04:24:87:CE:CB:DF:18:17:E2

X509v3 Subject Alternative Name:
  DNS:www.razick.com, DNS:www.razickZ.com, DNS:www.razickB.com, DNS:www.razickH.com, DNS:www.razickA.com
Signature Algorithm: sha256WithRSAEncryption
  97:7a:31:1e:b5:b9:bc:bb:00:7e:c6:85:75:fc:47:d0:16:c6:
  e0:98:78:30:56:ac:f1:18:1c:9e:cc:da:96:12:0b:85:c8:9b:
  11:3b:a9:db:01:4c:c4:1d:62:08:8f:b1:10:05:dd:a0:99:6b:
  19:57:32:1b:c5:57:fb:2a:6e:51:ec:93:3f:a2:6c:e0:91:7f:
  d2:25:c5:31:c3:5f:29:ba:86:f6:a1:e5:a2:59:44:af:4e:dc:
  c2:43:c5:ec:af:a9:15:d4:62:9b:d0:03:3b:d1:cc:13:9b:c9:
  83:e7:0e:68:ad:43:a5:24:c5:cf:33:22:94:bc:0c:76:3c:f1:
  e9:3f:7b:b3:86:20:f8:b9:e8:6f:dc:31:0e:f9:46:50:14:7b:
  77:04:b1:32:e0:75:a6:f1:6e:a3:91:58:d4:8f:ac:9d:ae:cf:
  b4:34:d5:61:76:23:48:9c:58:fc:c1:d6:7c:5a:5e:9a:5c:70:
  09:73:c1:f1:87:60:47:da:06:dd:3b:4e:db:6a:5b:c3:8d:49:
  2d:2c:ca:48:28:f8:3a:00:59:80:6e:f1:36:76:dd:2d:35:3f:
  ce:3f:00:cf:31:5c:c0:27:d4:40:89:6f:89:a8:b3:be:12:ab:
  7c:c8:90:96:43:40:cd:df:0b:7a:61:45:16:32:cf:8e:ce:63:
  a6:f6:5c:8d:e1:5a:76:d4:f2:20:cb:44:5f:4f:14:57:56:26:
  a9:2a:71:37:0b:88:0e:9c:f0:0a:98:22:04:79:fd:de:fb:21:
  69:8f:b2:69:16:ba:62:79:1b:b5:86:5c:ab:a4:88:a9:92:57:
  0a:aa:1f:86:9b:6a:28:68:44:eb:c2:de:4d:b9:9f:54:95:60:
  84:53:a2:87:b4:94:aa:fe:b2:d4:43:e5:73:1c:82:af:76:d6:
  0d:36:23:19:af:7f:93:2d:a7:83:68:44:d4:8c:ec:3e:19:ac:
  34:39:c5:d0:f0:bf:07:e1:c5:27:6c:2f:95:47:48:b5:3a:3d:
  6d:ee:16:7d:9d:4d:f2:7d:34:a2:ff:f5:86:b9:a2:92:7f:8c:
  1b:f4:41:6f:f7:97:39:43:eb:67:cb:ef:d1:8a:ea:87:a7:eb:
  6c:b1:92:b8:fc:d0:fc:51:8a:ce:84:4a:c7:15:17:67:ea:c1:
  91:cf:e2:bd:23:4a:ce:c1:1e:8a:3b:7c:85:e2:73:7b:3d:31:
  76:b6:fd:c0:3e:86:06:5e:4e:8f:8f:9f:22:be:d8:c4:8e:4b:
  2c:c8:62:ef:9a:b3:a5:2d:0e:52:57:d9:01:b6:ff:d8:bb:61:
  11:ae:99:c7:52:3b:d4:e4:8e:c3:b3:ef:f1:2b:b2:c0:65:3b:
  d5:9b:72:f0:17:bc:43:35
```

Task 4: Deploying Certificate in an Apache-Based HTTPS Website

The purpose of task 4 was to use the certificate that was generated from task 3 to point the browser to my web server. The first step was creating the Apache SSL Virtual Host configuration file. Given the file called *bank 32 apache ssl.conf*, I copied its contents into a new file called *razick_apache_ssl.conf* and modified the server name to match mine. Below is a screenshot of my configuration file, as well as a screenshot of the command I used to generate it.

```
GNU nano 4.8
<VirtualHost *:443>
    DocumentRoot /var/www/razick
    ServerName www.razick.com
    ServerAlias www.razickA.com
    ServerAlias www.razickB.com
    DirectoryIndex index.html

    SSLEngine On
    SSLCertificateFile /certs/server.crt
    SSLCertificateKeyFile /certs/server.key
</VirtualHost>
```

```
root@ac9a897b0cc0:/# nano /etc/apache2/sites-available/razick_apache_ssl.conf
```

Next, I created the web directory for my web server and created an *index.html* file so that my website would say *Hello World*. Below is a screenshot of the commands used for this.

```
root@ac9a897b0cc0:/certs# mkdir -p /var/www/razick
root@ac9a897b0cc0:/certs# echo '<!DOCTYPE html><html><body><h1>Hello World</h1></body></html>' > /var/www/razick/index.html
```

Next, I enabled the SSL module and enabled my site. Below is a screenshot of the commands used to do so.

```
root@ac9a897b0cc0:/certs# a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Module socache_shmcb already enabled
Module ssl already enabled
root@ac9a897b0cc0:/certs# a2ensite razick_apache_ssl
Enabling site razick_apache_ssl.
To activate the new configuration, you need to run:
  service apache2 reload
root@ac9a897b0cc0:/certs# service apache2 restart
* Restarting Apache httpd web server apache2
Enter passphrase for SSL/TLS keys for www.razick.com:443 (RSA):█
```

Next, I restarted my server using the following command.

```
root@ac9a897b0cc0:/certs# service apache2 restart [ OK ]
* Restarting Apache httpd web server apache2
Enter passphrase for SSL/TLS keys for www.razick.com:443 (RSA): [ OK ]
root@ac9a897b0cc0:/certs#
```

Following this, I tried to access my website and was met with the following error message: **“Hmm. We’re having trouble finding that site. We can’t connect to the server at www.razick.com. If that address is correct, here are three other things you can try: Try again later. Check your network connection. If you are connected but behind a firewall, check that Firefox has permission to access the Web.”** I realized that the issue was that my *hosts* file had not been correctly updated to map to the IP Address for my hosted website. Below is a screenshot of the *hosts* file, with the new line highlighted.

```
GNU nano 4.8 /etc/hosts
127.0.0.1 localhost
127.0.1.1 VM

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

# For DNS Rebinding Lab
192.168.60.80 www.seedIoT32.com

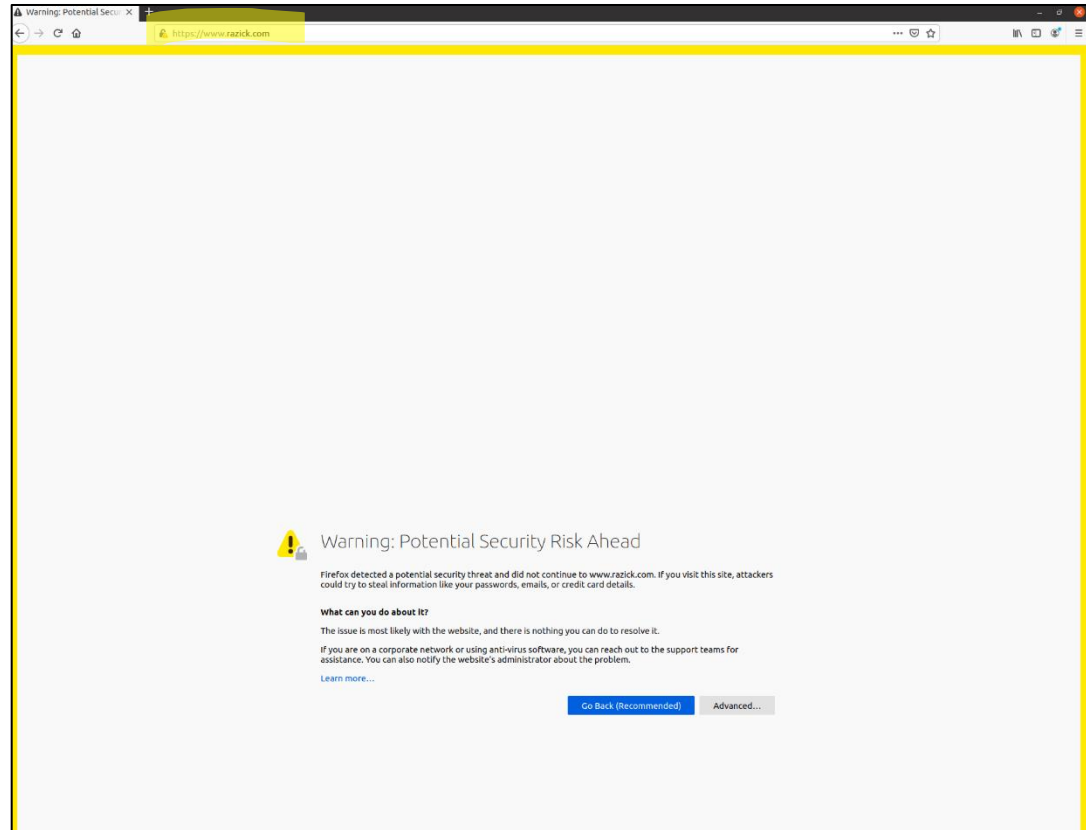
# For SQL Injection Lab
10.9.0.5 www.SeedLabSQLInjection.com

# For XSS Lab
10.9.0.5 www.xsslabelgg.com
10.9.0.5 www.example32a.com
10.9.0.5 www.example32b.com
10.9.0.5 www.example32c.com
10.9.0.5 www.example60.com
10.9.0.5 www.example70.com

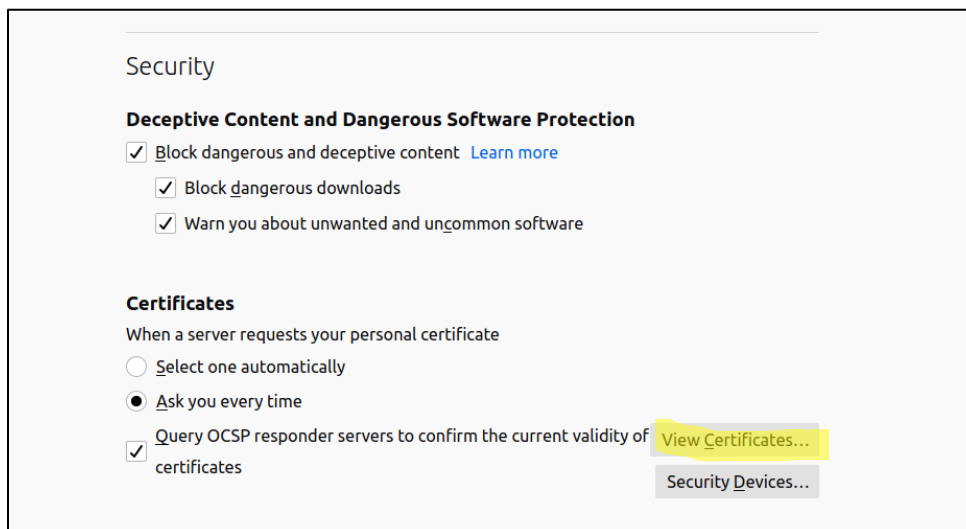
# For CSRF Lab
10.9.0.5 www.csrflabelgg.com
10.9.0.5 www.csrf-lab-defense.com
10.9.0.105 www.csrf-lab-attacker.com

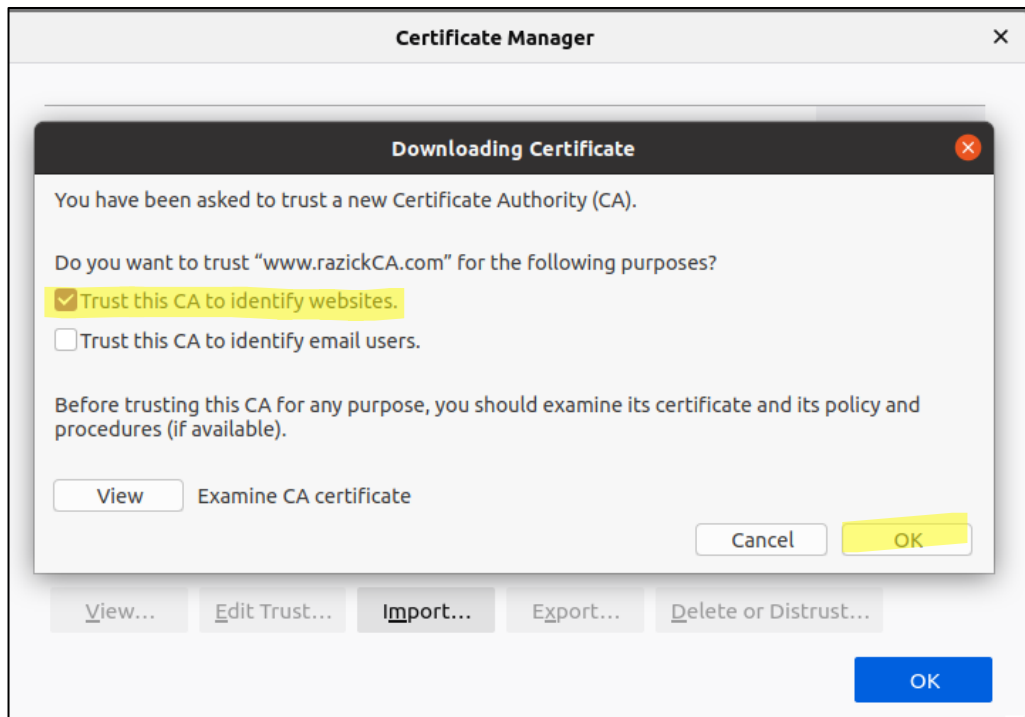
# For Shellshock Lab
10.9.0.80 www.seedlab-shellshock.com
10.9.0.80 www.razick.com
```

Now that Firefox could find my web server, I could browse my website. To the right is a screenshot of what I saw when I tried to browse my website. In order to solve this issue, I had to import my CA's certificate into Firefox as a trusted certificate authority.



Below are the screenshots depicting this process.





Now when I tried to browse my website, this is what I saw.

